

10-2 试求水平平行流过长度为 0.4 m 的平板时沿程 $x=0.1$ m、0.2 m、0.3 m、0.4 m 处的局部表面传热系数。已知水的来流温度 $t_\infty=20$ °C, 速度 $u_\infty=1$ m/s, 平板的壁面温度 $t_w=60$ °C。

$$\frac{h_x x}{\lambda} = 0.332 Re_x^{1/2} Pr^{1/3}$$

$$h_x = 0.332 Re_x^{1/2} Pr^{1/3} \left(\frac{\lambda}{x} \right)$$

等壁温平板层流换热、定性温度:

$$t_m = \frac{1}{2}(t_w + t_\infty) \\ = \frac{1}{2}(60 + 20) = 40^\circ\text{C}$$

根据附录18

$$t = 40^\circ\text{C} \quad \nu = 0.659 \times 10^{-6} \text{ m}^2/\text{s} \quad \lambda = 63.5 \times 10^{-2} \text{ W/m}\cdot\text{K} \\ Pr = 4.31$$

$$Re_x = \frac{u_\infty x}{\nu} = \frac{1}{\nu} x = 1.5175 \times 10^6 x$$

$$h_{0.1} = 0.332 [(1.5175 \times 10^6)(0.1)]^{1/2} (4.31)^{1/3} \left(\frac{0.635}{0.1} \right) = 1336.498 \text{ W/m}^2\cdot\text{K}$$

$$h_{0.2} = 0.332 [(1.5175 \times 10^6)(0.2)]^{1/2} (4.31)^{1/3} \left(\frac{0.635}{0.2} \right) = 945.0474 \text{ W/m}^2\cdot\text{K}$$

$$h_{0.3} = 0.332 [(1.5175 \times 10^6)(0.3)]^{1/2} (4.31)^{1/3} \left(\frac{0.635}{0.3} \right) = 771.628 \text{ W/m}^2\cdot\text{K}$$

$$h_{0.4} = 0.332 [(1.5175 \times 10^6)(0.4)]^{1/2} (4.31)^{1/3} \left(\frac{0.635}{0.4} \right) = 668.2494 \text{ W/m}^2\cdot\text{K}$$

10-3 如果将上题中的水改为空气, 其他参数保持不变, 试计算整个平板的平均表面传热系数以及单位宽度平板的换热量, 并对比这两种情况的计算结果。

由附表16可得:

$$t = 40^\circ\text{C} \quad \lambda = 0.0276 \text{ W/m}\cdot\text{K} \quad \nu = 16.96 \times 10^{-6} \text{ m}^2/\text{s} \quad , \quad Pr = 0.699$$

$$Re = \frac{u_\infty x}{\nu} = \frac{1(0.4)}{16.96 \times 10^{-6}}$$

$$h_{0.4} = 0.332 Re^{1/2} Pr^{1/3} \frac{\lambda}{x} = 0.332 \left(\frac{0.4}{16.96 \times 10^{-6}} \right)^{1/2} (0.699)^{1/3} \left(\frac{0.0276}{0.4} \right)$$

$$h_{0.4} = 3.1222 \text{ W/m}^2\cdot\text{K}$$

$$h = 2h_{0.4} = 2h_{0.4} = 6.2444 \text{ W/m}^2\cdot\text{K}$$

$$\Phi = Ah(t_w - t_\infty)$$

$$\text{单位宽度} \Rightarrow A = 1 \times 0.4 = 0.4 \text{ m}^2$$

$$\Phi = 0.4(6.2444)(60 - 20) = 99.9104 \text{ W}$$

空气局部和平均传热系数均小于水, 故换热量自然小于水。

10-4 如果用特征长度为原型 1/5 的模型来模拟原型中速度为 6 m/s、温度为 200 °C 的空气强迫对流换热, 模型中空气的温度为 20 °C。试问模型中空气的速度应为多少? 如果测得模型中对流换热的平均表面传热系数为 200 W/(m² · K), 求原型中的平均表面传热系数数值。

$$t_\infty = 200^\circ\text{C} \quad \lambda = 0.0393 \text{ W/m}\cdot\text{K} \quad \nu = 34.85 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr = 0.68$$

$$t_\infty' = 20^\circ\text{C} \quad \lambda' = 0.0257 \text{ W/m}\cdot\text{K} \quad \nu' = 15.06 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr' = 0.703$$

$$Re = Re' \quad \frac{u l}{\nu} = \frac{u' l'}{\nu'} \Rightarrow u' = \frac{u l \nu'}{l' \nu} = (5) \left(\frac{6(15.06)}{34.85} \right) = 12.9641 \text{ m/s}$$

$$h = 200 \text{ W/m}^2\cdot\text{K}$$

$$Nu = Nu' \quad \frac{h l}{\lambda} = \frac{h' l'}{\lambda'} \Rightarrow h = h' \frac{l'}{l} \left(\frac{\lambda}{\lambda'} \right) = 60.695 \text{ W/m}^2\cdot\text{K}$$

10-5 水在换热器管内被加热,管内径为 14 mm,管长为 2.5 m,管壁温度保持为 110 °C,水的进口温度为 50 °C,流速为 1.3 m/s,试求水通过换热器后的温度。

$$\Delta t = \frac{\Delta t' - \Delta t''}{\ln \frac{\Delta t'}{\Delta t''}}$$

$$\Delta t' = 110 - 50 = 60^\circ\text{C}$$

设出口温度为 94 °C

$$\Delta t'' = 16^\circ\text{C}$$

$$\Delta t = 33.29^\circ\text{C}$$

$$t_f = t_w - \Delta t = 76.71^\circ\text{C}$$

$$\rho = 977.7 - \frac{76.71 - 70}{80 - 70} (977.7 - 971.8) = 973.7411 \text{ kg/m}^3$$

$$\lambda = 60.8 + \frac{76.71 - 70}{80 - 70} (67.4 - 66.8) = 67.2026 \times 10^{-2} \text{ W/m}\cdot\text{K}$$

$$\nu = 0.418 - \frac{76.71 - 70}{80 - 70} (0.418 - 0.365) = 0.38145 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr = 2.55 - \frac{76.71 - 70}{80 - 70} (2.55 - 2.21) = 2.32186$$

$$\eta_f = 406.1 - \frac{76.71 - 70}{80 - 70} (406.1 - 355.1) = 371.879 \times 10^{-6} \text{ kg/m}\cdot\text{s}$$

$$C_p = 4.187 + \frac{76.71 - 70}{80 - 70} (4.195 - 4.187) = 4.1924 \text{ kJ/kg}\cdot\text{K}$$

$$Re_f = \frac{u \cdot d}{\nu} = \frac{1.3(0.014)}{0.38145 \times 10^{-6}} = 4.77127 \times 10^4 \Rightarrow \text{紊流}$$

$$Nu_f = 0.027 Re_f^{0.8} Pr_f^{\frac{1}{4}} \left(\frac{\eta_f}{\eta_w} \right)^{0.14} \quad t_w = 110^\circ\text{C} \quad \eta_w = 259 \times 10^{-6} \text{ kg/m}\cdot\text{s}$$

$$Nu_f = 208.0705$$

$$h = Nu_f \frac{\lambda}{d} = 9987.7717 \text{ W/m}^2\cdot\text{K}$$

$$27.6806$$

$$\Phi_1 = h F \Delta t \quad F = \pi d l = 0.11 \text{ m}^2$$

$$\Phi_1 = 36574.2212 \text{ W}$$

$$\Phi_2 = \rho V A C_p (94 - 80) = 25944.3 \text{ W}$$

$\Phi_1 \approx \Phi_2$ 出口温度为 94°C 最相近

10-6 用内径为 0.016 m、长为 2.5 m 的不锈钢管进行管内对流换热实验，实验时直接对不锈钢管通以直流电加热，电压为 5 V，电流为 900 A，水的进口温度为 20°C ，流速为 0.5 m/s，管外用保温材料保温，忽略热损失。试求管内对流换热的表面传热系数及换热温差。

$$\Phi = VI = 4500 \text{ W}$$

$$\Phi = h \pi d L \Delta T = \rho V A C_p (t_{f''} - t_{f'})$$

$$t_f = \frac{1}{2} (t_{f''} + t_{f'})$$

$$t_{f'} = 20^\circ\text{C}$$

$$\rho = 998.2 \text{ kg/m}^3 \quad C_p = 4.183 \text{ kJ/kg}\cdot\text{K}$$

$$\lambda = 59.9 \times 10^{-2} \text{ W/m}\cdot\text{K} \quad \eta = 1004 \times 10^{-6} \text{ kg/m}\cdot\text{s} \quad \nu = 1.006 \times 10^{-6} \text{ m}^2/\text{s}$$

$$Pr = 7.02$$

$$4500 = 998.2 (0.5) (4.183) \left(\frac{\pi (0.016)^2}{4} \right) (t_{f''} - 20)$$

$$t_{f''} = 30.72^\circ\text{C}$$

$$t_f = 25.36^\circ\text{C}$$

$$Pr = 6.16 \quad \nu = 0.8983 \times 10^{-6} \text{ m}^2/\text{s} \quad \eta = 895.46 \times 10^{-6} \text{ kg/m}\cdot\text{s} \quad \lambda = 60.9184 \times 10^{-2} \text{ W/m}\cdot\text{K}$$

$$C_p = 4.178 \text{ kJ/kg}\cdot\text{K} \quad \rho = 996.81 \text{ kg/m}^3$$

$$Re_f = \frac{u_{\infty} d}{\nu} = 8905.7 \quad \text{过渡区}$$

$$Nu_f = 0.012 (Re_f^{0.87} - 280) Pr_f^{0.4} \left[1 + \left(\frac{d}{L} \right)^{2/3} \right] \left(\frac{Pr_f}{Pr_w} \right)^{0.11}$$

$$0.05 < \frac{Pr_f}{Pr_w} < 20 \quad \Delta t = 14^\circ\text{C}, \quad t_w = 31.36^\circ\text{C}$$

$$Pr_w = 4.38 \quad \frac{Pr_f}{Pr_w} = 1.416 \quad \text{满足适用条件}$$

$$N_{uf} = 65.35$$

$$h = N_{uf} \frac{\lambda}{d} = 2488.125 \text{ W/m}^2 \cdot \text{K}$$

$$Q = h \pi d L \Delta t = 4500 \quad \Delta t = 14.4^\circ\text{C} \approx 14^\circ\text{C}$$

10-7 空气以 1.3 m/s 速度在内径为 22 mm、长为 2.25 m 的管内流动，空气的平均温度为 38.5 °C，管壁温度为 58 °C，试求管内对流换热的表面传热系数。

$$t_w = 58^\circ\text{C} \quad t_f = 38.5^\circ\text{C}$$

$$\rho = 1.1336 \text{ kg/m}^3 \quad C_p = 1.005 \text{ kJ/kg} \cdot \text{K} \quad \lambda = 2.7465 \times 10^{-2} \text{ W/m} \cdot \text{K}$$

$$\mu = \eta = \nu \rho$$

$$\mu_f = 19.025 \times 10^{-6} \text{ kg/m} \cdot \text{s} \quad \nu = 16.816 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr = 0.2993$$

$$\mu_w = 20 \times 10^{-6} \text{ kg/m} \cdot \text{s}$$

$$Re_f = \frac{u_{\infty} d}{\nu} = 1700.8 < 2300 \quad \text{层流}$$

$$N_{uf} = 1.86 (Re_f Pr_f \frac{d}{L})^{\frac{1}{3}} \left(\frac{\mu_f}{\mu_w} \right)^{0.14} = 4.18$$

$$h = N_{uf} \frac{\lambda}{d} = 5.224 \text{ W/m}^2 \cdot \text{K}$$

10-10 一套管式换热器，内管外径 $d_1 = 12 \text{ mm}$ ，外管内径 $d_2 = 16 \text{ mm}$ ，管长为 400 mm，在内外管之间的环形通道内水的流速 $u = 2.4 \text{ m/s}$ 、平均温度 $t_f = 73^\circ\text{C}$ ，内管壁温 $t_w = 96^\circ\text{C}$ ，试求内管外表面处对流换热的表面传热系数。

$$\rho = 975.93 \text{ kg/m}^3 \quad C_p = 4.1814 \text{ kJ/kg} \cdot \text{K} \quad \lambda = 66.98 \times 10^{-2} \text{ W/m} \cdot \text{K}$$

$$\eta_f = 310.8 \times 10^{-6} \text{ kg/m} \cdot \text{s} \quad \nu = 0.4 \times 10^{-6} \text{ m}^2/\text{s} \quad Pr = 2.448$$

$$\eta_w = 295.46 \times 10^{-6} \text{ kg/m} \cdot \text{s}$$

$$Re_f = \frac{u_{\infty} d_r}{\nu} \quad d_r = \frac{4A_c}{p} = \frac{\pi(d_2^2 - d_1^2)}{\pi(d_2 + d_1)} = d_2 - d_1$$

$$Re_f = 24000 > 10^4 \Rightarrow \text{紊流}$$

$\Delta t = 23^\circ\text{C} < 50^\circ\text{C}$ 气体 $\Rightarrow$ 紊流 Dittus-Boelter

$$Nu_f = 0.023 Re_f^{0.8} Pr_f^n \quad t_w > t_f \Rightarrow n = 0.4$$

$$Nu_f = 105.086$$

$$h_i = Nu_f \frac{\lambda}{d} = 5.863883 \times 10^5 \text{ W/m}^2\cdot\text{K}$$